

Development of a Liquid Crystal Display Spatial Light Modulator for Optical Applications

THESIS

Presented to the Faculty of the Department of Physics and Astronomy
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PHYSICS AND ASTRONOMY

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ABSTRACT

This thesis investigates the use of a low-cost consumer liquid crystal display (LCD) as a transmissive spatial light modulator (SLM) and quantitatively characterizes its optical performance using the Transport-of-Intensity Equation (TIE). A commercial LCD was modified into an SLM, integrated into a laser-based optical system, and controlled with a microcontroller for pixel-level modulation. Phase information was reconstructed from defocused intensity measurements using a numerical TIE implementation in MATLAB. The resulting phase and effective surface profiles reveal nonuniformities and defects that limit modulation fidelity. This work demonstrates the feasibility of low-cost SLM development and highlights the utility of TIE-based phase reconstruction for surface characterization, defect identification, and accessible optical metrology.

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I. Introduction

Liquid crystal displays (LCDs) are widely used in everyday technologies such as smartphones, laptops, and televisions. Beyond their role in visual display, LCDs are optical devices capable of manipulating light through pixel-level control of liquid crystal orientation [1]. This property allows LCDs to function as spatial light modulators (SLMs), devices that can modify the amplitude and/or phase of an incident laser beam to generate structured optical fields [2].

Commercial SLMs provide high optical quality and precise phase control, but their high-cost limits accessibility for small laboratories and educational settings. As a result, low-cost consumer LCDs represent an attractive alternative. However, their optical performance, particularly phase behavior, surface uniformity, and pixel-level modulation fidelity, is not well characterized for precision optical applications [1]. These limitations can reduce image resolution, introduce aberrations, and lead to unstable modulation, motivating systematic investigation.

The purpose of this project is to develop and evaluate a low-cost LCD-based spatial light modulator and to characterize its optical performance using image-based testing and phase reconstruction techniques. The central hypothesis is that image quality limitations in LCD-based SLMs are linked to phase nonuniformities and surface irregularities in the LCD. To test this hypothesis, the Transport-of-Intensity Equation (TIE) will be applied to reconstruct the optical phase of the SLM from a series of defocused intensity images. TIE is a computational imaging method commonly used for three-dimensional (3D) phase and surface reconstruction without interferometry [3]. By combining spatial light modulation experiments with TIE-based phase

analysis, this work aims to improve understanding of low-cost optical modulation systems and support more accessible platforms for optics education and research.

II. Methods

During this semester, I worked on constructing a three-dimensional (3D) image using the Transport of Intensity Equation (TIE) method to reconstruct the surface structure of a liquid crystal display (LCD) material. The Transport-of-Intensity Equation relates how measured light intensity changes along the direction of propagation to spatial variations in optical phase. By capturing intensity images at slightly different focus positions, the phase distribution of the spatial light modulator (SLM) surface can be reconstructed without the use of an interferometer.

This experiment relies solely on intensity measurements and allows the textures and structural variations of the material to be visualized quantitatively. The purpose of this experiment is to reconstruct the phase distribution of a light wave after it passes through (or reflects from) a sample and then convert that phase into a three-dimensional surface height map. A critical concept is that conventional cameras measure only intensity, not phase. The TIE method enables indirect recovery of phase information from measurable intensity variations.

The Transport-of-Intensity Equation is expressed as:

$$\frac{\partial I(x, y, z)}{\partial z} = -\frac{\lambda}{2\pi} \nabla_{\perp} \cdot [I(x, y, z) \nabla_{\perp} \phi(x, y, z)]$$

where $I(x, y, z)$ is the intensity, $\phi(x, y, z)$ is the optical phase, λ is the wavelength, and $\nabla_{\perp}$ represents the transverse gradient. The left side of the equation represents the axial intensity derivative, or how the intensity changes when the focus position is slightly shifted. The right side relates this change to the curvature of the wavefront phase.

LCD Preparation and Control System

A consumer-grade LCD was repurposed to function as a spatial light modulator. The display consists of multiple layers, including the liquid crystal medium, glass substrates, polarizers, backlight unit, electrodes, and driver circuitry. To access the active modulation layer, the LCD was carefully disassembled by peeling away outer components such as the casing and backlight while preserving the liquid crystal layer and electrode structure. As shown in Figure 1.

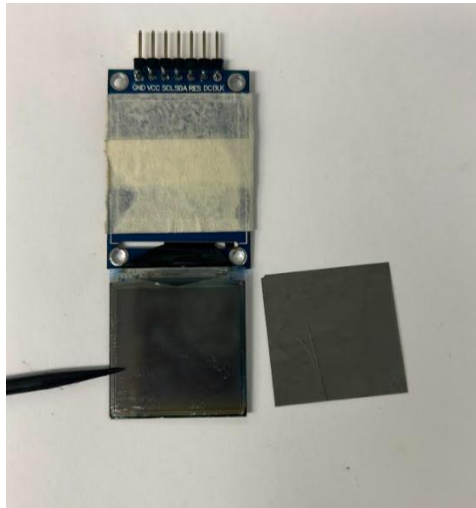


Figure 1 shows the LCD display after removal of diffuser and backlight layers, exposing the liquid crystal layer used for transmissive spatial light modulation.

Electrical connections were established by soldering conductive leads to the exposed contact pads of the LCD using the original metal connectors. These leads were interfaced with a ESP32 microcontroller. Cleanliness during this process was critical, as contamination or poor soldering resulted in unreliable connections and signal degradation.

The LCD was driven using custom C++ code implemented through the Arduino development environment. The microcontroller was programmed to display a sequence of predefined patterns, including uniform (no-pattern) states, on the LCD. These patterns were displayed in a controlled

temporal sequence, synchronized with image acquisition from the camera. This synchronization enabled systematic capture of intensity data under varying modulation conditions for subsequent phase reconstruction.

Experimental set-up

The optical system consisted of a laser source, three optical lenses, a spatial light modulator (LCD sample), and a camera mounted along an optical rail. Lens 1 was positioned 1.5 cm from the laser (65 cm on the ruler). Lens 2 was placed directly adjacent to Lens 1. Lens 3, which served as the imaging lens, was located 34 cm away (112 cm on the ruler), and the camera was positioned 20 cm from Lens 3 (132 cm on the ruler). This arrangement is shown in Figure 2.

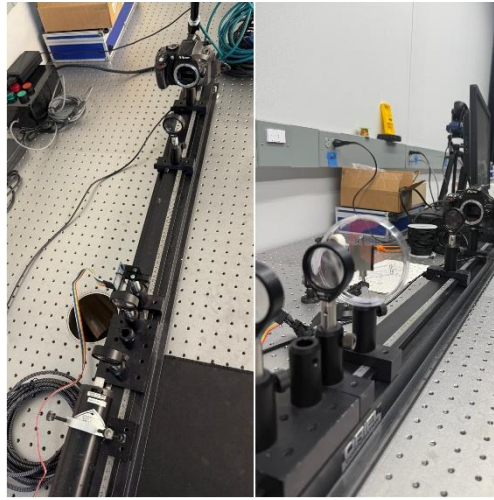


Figure 2 Optical setup used for the experiment. (a) Optical rail with three lenses, a camera, and the LCD connected for spatial light modulation testing. (b) LCD sample positioned in the optical path for Transport-of-Intensity Equation (TIE) measurements.

The LCD material was mounted carefully in front of the imaging lens to maintain consistent positioning within the camera frame. Maintaining alignment between images is critical, as even slight shifts can introduce artificial intensity variations that distort the reconstruction.

The imaging lens moved slightly forward (z^+) and backward (z^-) relative to the focal plane. The axial intensity derivative was approximated using the central difference formula:

$$\frac{\partial I}{\partial z} \approx \frac{I_+ - I_-}{2dz}$$

Numerical Reconstruction Method

After computing the axial intensity derivative, the next step was to rewrite the TIE in Poisson form:

$$\nabla^2 \phi(x, y) = -\frac{2\pi}{\lambda} \frac{1}{I_0(x, y)} \frac{\partial I(x, y, z)}{\partial z}.$$

This equation relates the Laplacian (curvature) of the phase to experimentally measured intensity changes. The captured intensity data are therefore converted into a Poisson equation whose solution yields the spatial phase distribution.

To solve this efficiently, the equation was transformed into the Fourier domain using the Fast Fourier Transform (FFT). In Fourier space, the Laplacian operator becomes multiplication by $-k^2$, leading to:

$$\Phi(k_x, k_y) = \frac{\text{RHS}(k_x, k_y)}{-k^2}$$

An inverse FFT was then applied to recover the spatial phase map. The zero-frequency term ($k = 0$) was explicitly set to zero to avoid division by zero and eliminate artificial global curvature. Additionally, planar tilt was fitted and subtracted from the recovered phase to remove low-frequency illumination gradients and global slope that do not represent true surface structure.

The recovered phase was then converted into physical surface height using:

$$h(x, y) = \frac{\lambda}{4\pi} \phi(x, y),$$

This relation converts optical phase shift into height variation for reflection geometry. The result is a quantitative three-dimensional surface map derived entirely from intensity measurements.[4]

III. Results

To evaluate the effectiveness of the Transport of Intensity Equation (TIE) reconstruction, the processed intensity derivatives, recovered phase distributions, and corresponding surface height maps are presented in Figures 5–32. These results demonstrate how patterns generated by the display code propagate through the reconstruction process and appear in the resulting 3D profiles. Notably, the reconstructed surfaces exhibit apparent depth variations that closely follow the imposed intensity patterns, even in cases where no physical height variation is expected. This behavior suggests that the TIE algorithm is highly sensitive to spatial intensity gradients introduced by the display. Because TIE relates axial intensity variations to phase, structured illumination can be misinterpreted as phase curvature, leading to artificial depth features in the reconstructed surface. This highlights a key limitation of TIE-based reconstruction under non-uniform illumination conditions.

To emulate spatial light modulation, an Arduino-controlled LCD was used to generate programmable intensity patterns, analogous to beam shaping in a spatial light modulator (SLM). In this implementation, patterns were created using C++ code to systematically vary the spatial distribution of light intensity when illuminated by a laser source. Figure 1 presents representative examples of these generated patterns prior to their use in the TIE reconstruction process. The

patterns exhibit variations in shape, size, and spatial frequency, providing controlled test conditions for evaluating the response of the reconstruction algorithm. The goal is to determine whether the TIE reconstruction can accurately reproduce these imposed intensity variations as corresponding features in the recovered phase and 3D surface profiles.

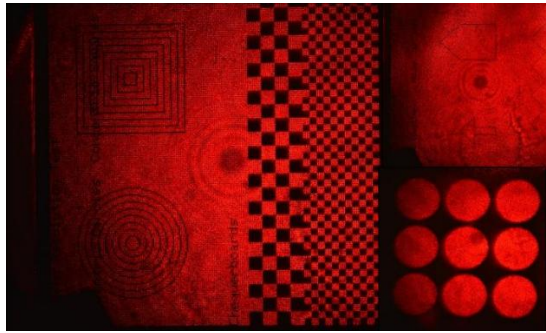


Figure 3 shows examples of spatial light modulation patterns generated using the Arduino-controlled LCD.

March 2026 results

For the initial TIE reconstruction, the LCD was used in an unmodulated state, with no applied Arduino control signal, effectively representing the SLM in an “off” condition. In this case, the LCD surface itself including inherent defects and structural features served as the source of intensity variation. Shown in figures 2-7.

The reconstruction reveals several consistent features of the physical LCD. A prominent defect located near the bottom of the screen, likely a scratch or dent, is clearly reproduced in the 3D surface profile. Additionally, diagonal line structures observed on the LCD surface are also

visible in the reconstructed image, particularly in the upper region, indicating that the TIE method is capable of capturing real surface characteristics under these conditions.

However, the overall reconstruction exhibits significant irregularities, appearing lumpy and wave-like rather than smooth. This behavior is consistent with earlier trials in which similar wave-like artifacts were observed. These distortions are likely due to noise in the system, including environmental factors such as air fluctuations, particle interactions, and instability in illumination when the SLM is not actively controlled. To mitigate these effects, noise reduction techniques were implemented in the MATLAB reconstruction code; however, residual artifacts remain, suggesting that noise continues to influence the recovered phase and height profiles. This suggests that while TIE can recover true surface features, its accuracy is strongly affected by noise when intensity modulation is not well controlled.

Broken LCD (SLM Off)

Trial 1

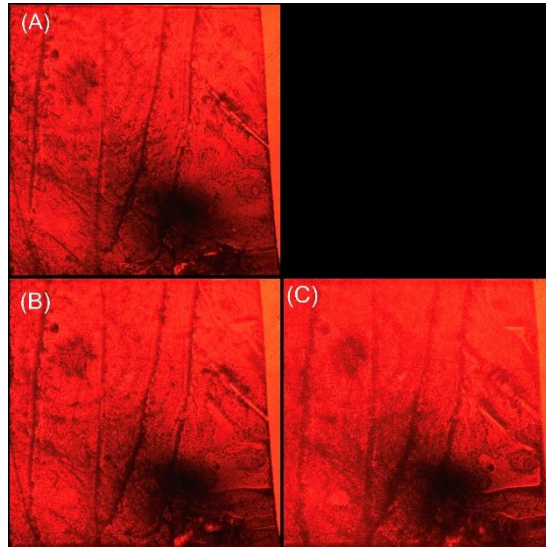


Figure 4 shows an LCD scrap from peeling. (A) Primary image (B) 5mm ahead of I0 (c) 5mm behind I0

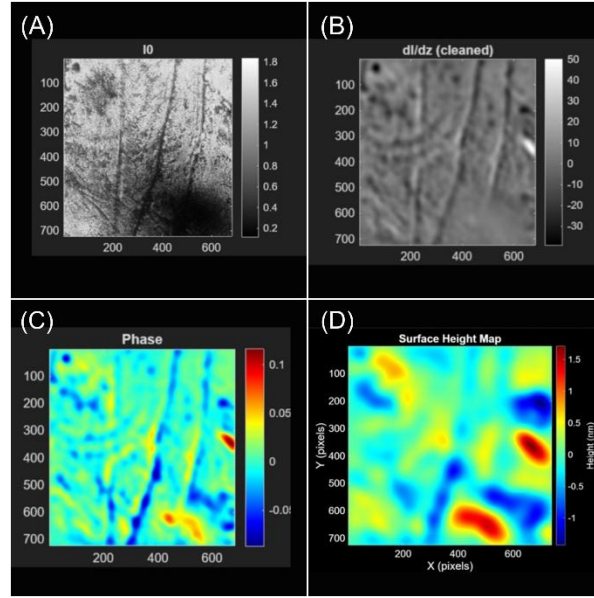


Figure 5 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

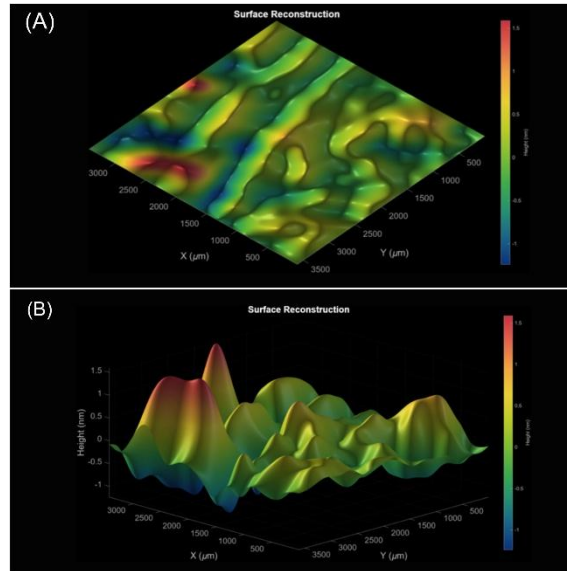


Figure 6 shows the 3d reconstruction of the LCD SLM turned off and shows characteristics of the damage. (A) top view (B) side view

Trial 2

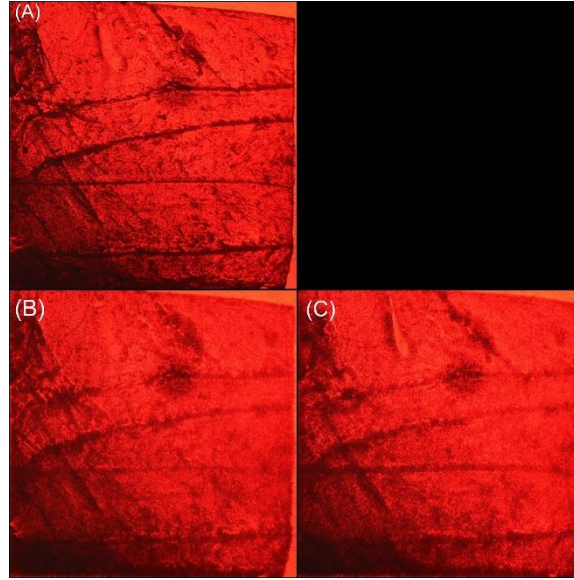


Figure 7 shows an LCD scrap from peeling. (A) Primary image (B) 5mm ahead of I_0 (c) 5mm behind I_0

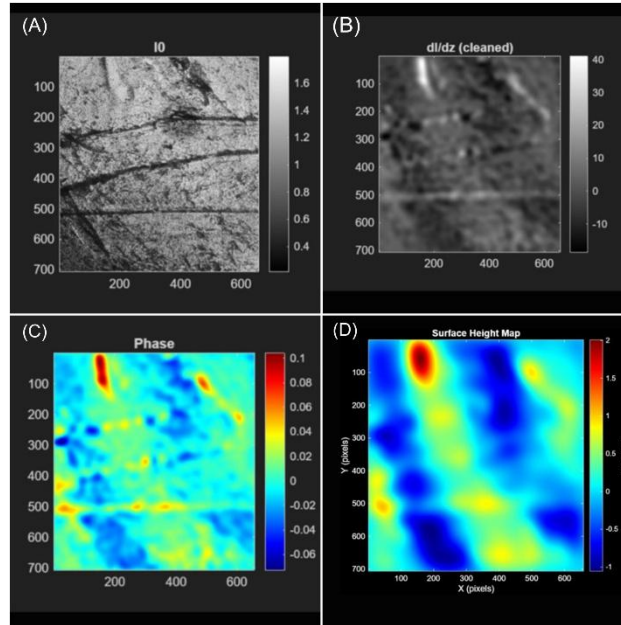


Figure 8 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

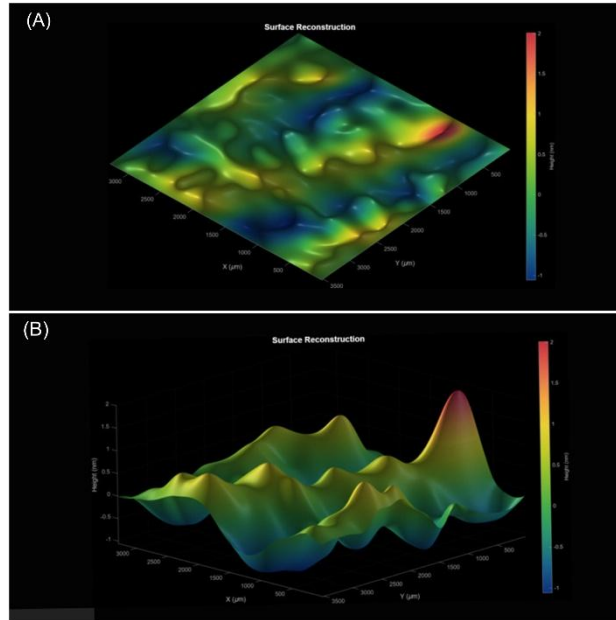


Figure 9 shows the 3d reconstruction of the LCD SLM turned off and shows characteristics of the damage. (A) top view (B) side view

Normal LCD (SLM On)

The TIE reconstructions were next performed with the SLM actively modulated using Arduino-generated patterns. Figures 10- 33 present TIE reconstructions obtained under patterned illumination for both intact and damaged LCD conditions. Despite differences in the specific patterns applied, all reconstructions exhibit consistent behavior: the imposed intensity patterns introduce structured variations in the recovered phase and surface height maps. In the case of the intact LCD (Figures 8-10), periodic features corresponding to the projected patterns are clearly visible, even in regions where no physical surface variation is present. This indicates that the TIE algorithm is interpreting intensity modulation as phase structure, resulting in artificial surface features.

For the damaged LCD (Figures 13-33), these effects are further amplified. Existing defects interact with the imposed patterns, producing exaggerated peaks and valleys in the reconstructed surface. As a result, defects appear significantly more pronounced compared to the unmodulated case, demonstrating that structured illumination enhances both real and artificial features.

Additionally, variations in pattern spatial frequency influence the appearance of the reconstruction. Higher-frequency patterns produce more localized and rapidly varying features, while lower-frequency patterns result in smoother, wave-like structures. These results confirm that the observed depth variations are not solely due to physical surface geometry but are strongly influenced by intensity gradients introduced by the display.

Normal LCD (SLM On)

Checkered board circles and squares

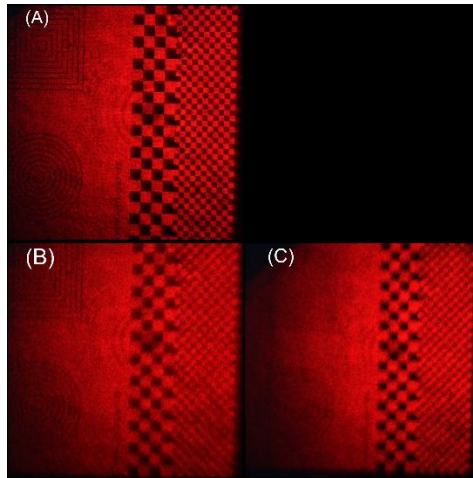


Figure 10 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Squares on the right-side pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z).

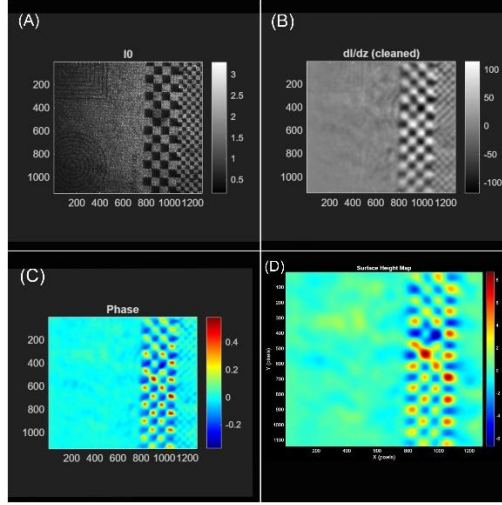


Figure 11 (A) shows the in-focus intensity I_0 , serving as the reference image with surface texture and illumination variations. (B) shows the axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) shows the recovered phase distribution, highlighting optical path variations in the LCD. Planar tilt was removed to isolate structural features, with minor noise present in low-intensity regions due to small I_0 values. (D) presents the corresponding two-dimensional height map.

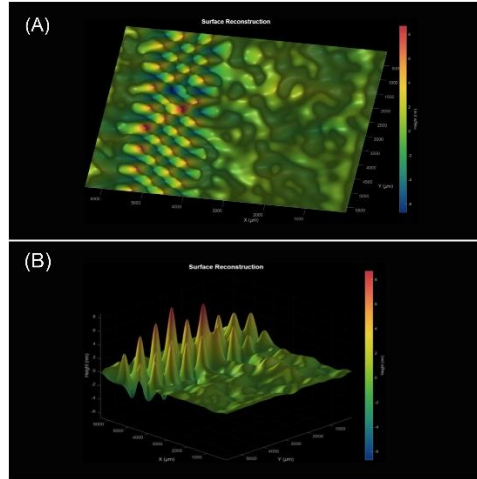


Figure 12 displays the three-dimensional surface reconstruction of a normal LCD with the SLM displaying a pattern, shown (B) side and (A) top views. Peaks and valleys correspond to texture features in the original intensity images.

Broken LCD (SLM On)

Checkered board circles and squares

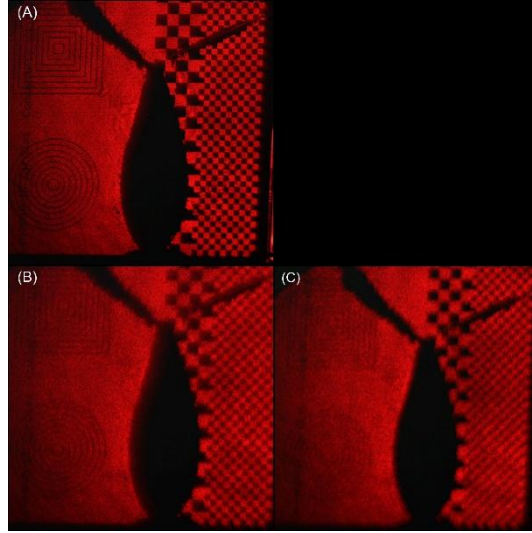


Figure 13 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a checkerboard pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

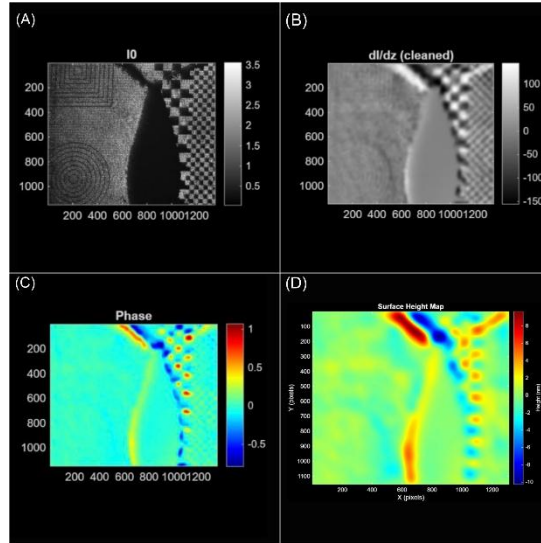


Figure 14 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

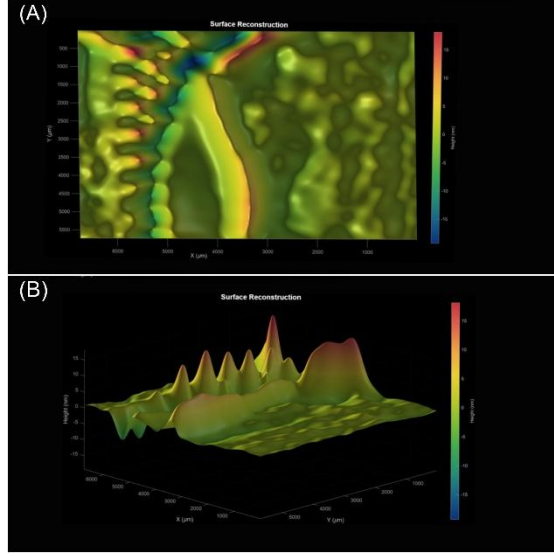


Figure 15 shows the 3d reconstruction of the checked damaged LCD. The damages show to be more intense than the checked patterns. (A) top view (B) Side view

Squares on the right side

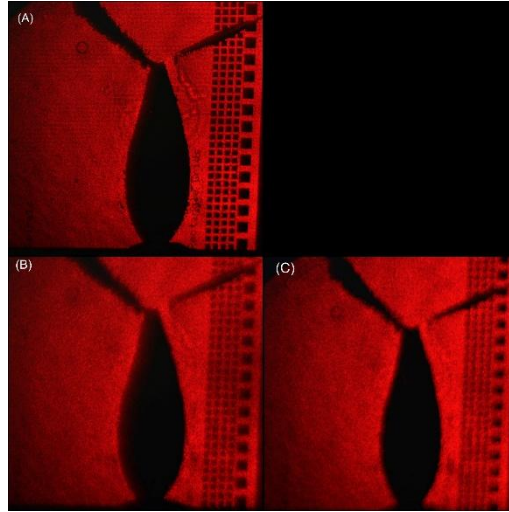


Figure 16 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Squares on the right-side pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

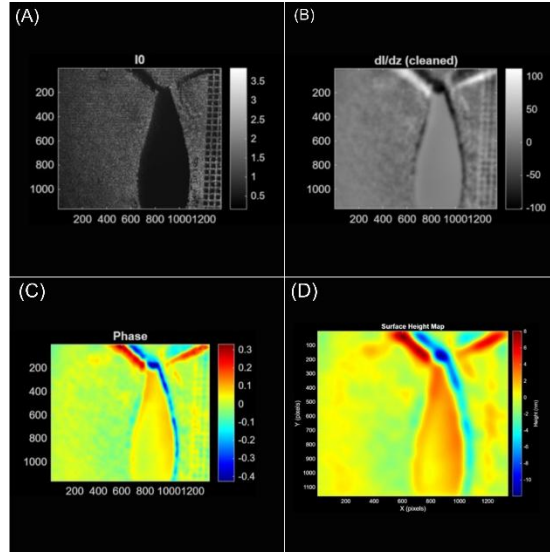


Figure 17 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

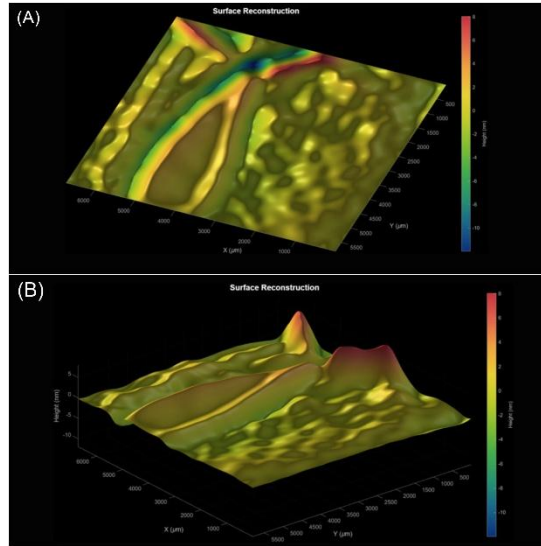


Figure 18 shows the 3d reconstruction of the checked damaged LCD. The damages show to be more intense than the checked patterns. (A) top view (B) Side view

April 2026 Results

Additional patterns were generated to replicate common spatial light modulator (SLM) outputs, including diffractive structures such as concentric circles and line gratings. These patterns were used to further evaluate the response of the TIE reconstruction under structured illumination.

The resulting reconstructions are consistent with the corresponding in-focus intensity images I_0 , with the imposed patterns clearly reflected in the recovered phase and surface height maps. This agreement indicates that the TIE algorithm effectively captures the spatial distribution of intensity variations introduced by the display.

Broken LCD (SLM On)

Stripes and Interference circle waves

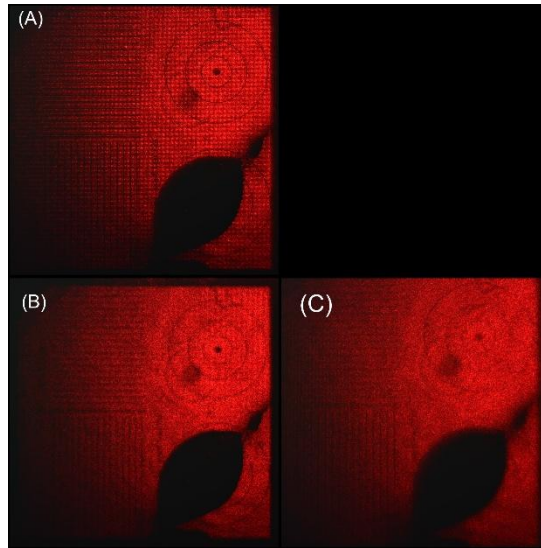


Figure 19 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Stripes and interference circle waves pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

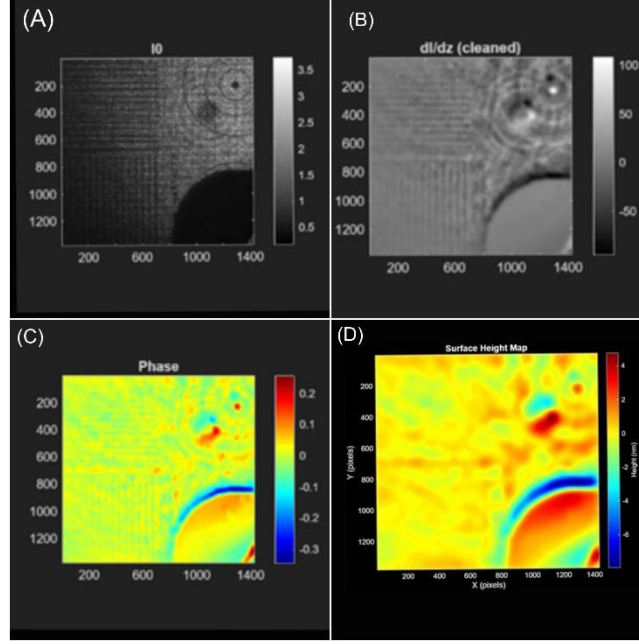


Figure 20 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

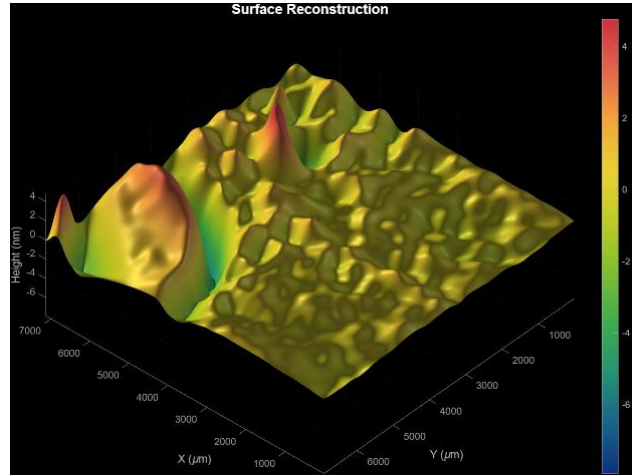


Figure 21 shows the three-dimensional reconstruction of the projected pattern. The damaged region of the LCD appears more pronounced than the imposed stripe and circular pattern, indicating that surface defects produce stronger intensity variations and dominate the TIE reconstruction.

Stripes

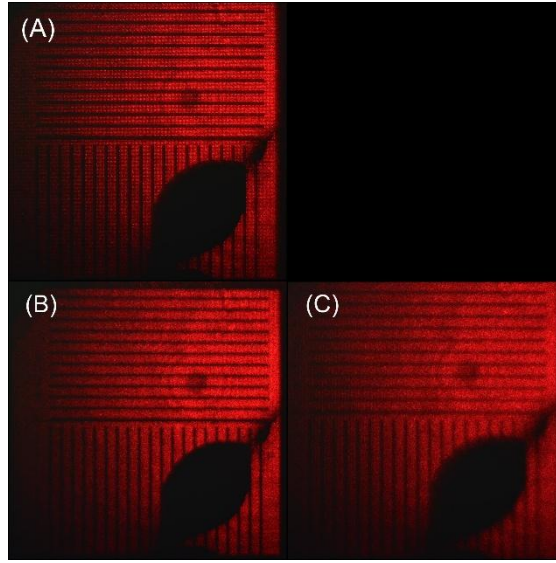


Figure 22 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a Stripes pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

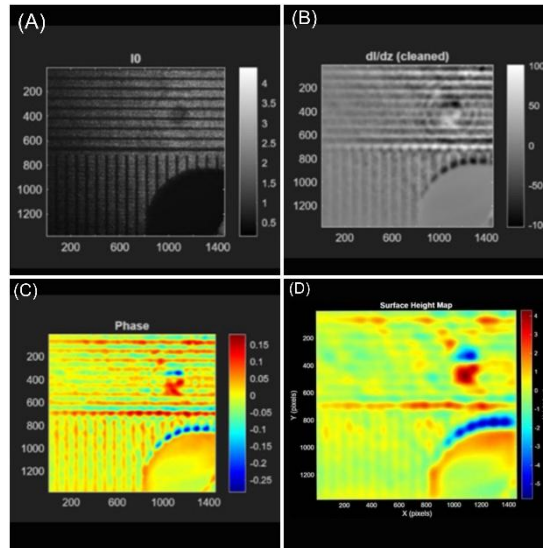


Figure 23 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

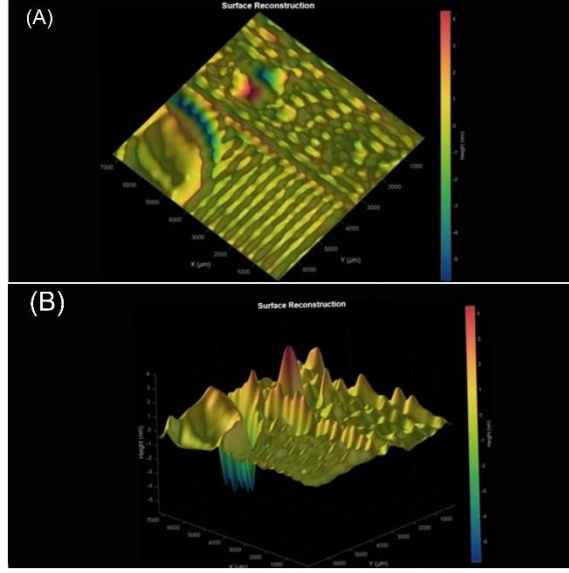


Figure 24 shows the three-dimensional reconstruction of the stripe pattern. The reconstructed features are consistent with the reference image I_0 , with the stripe structure clearly preserved. Additionally, characteristics of the damaged LCD, including surface irregularities, are visible in the reconstruction.

Interference circle waves

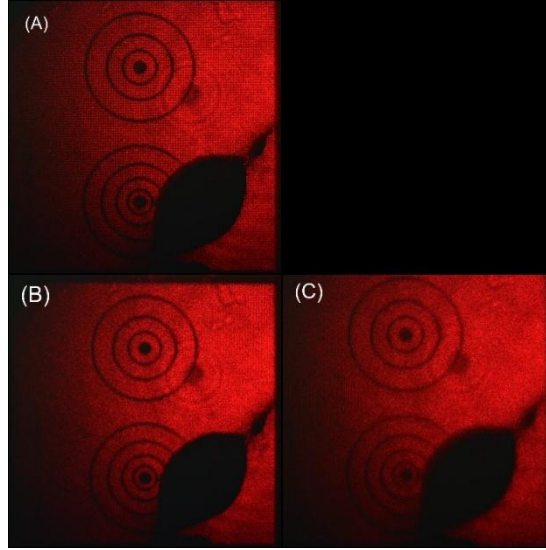


Figure 25 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying an interference circle wave pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

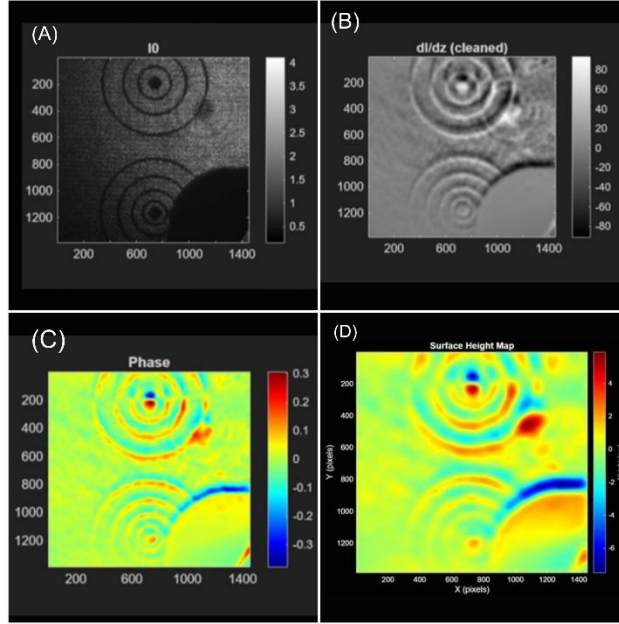


Figure 26 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

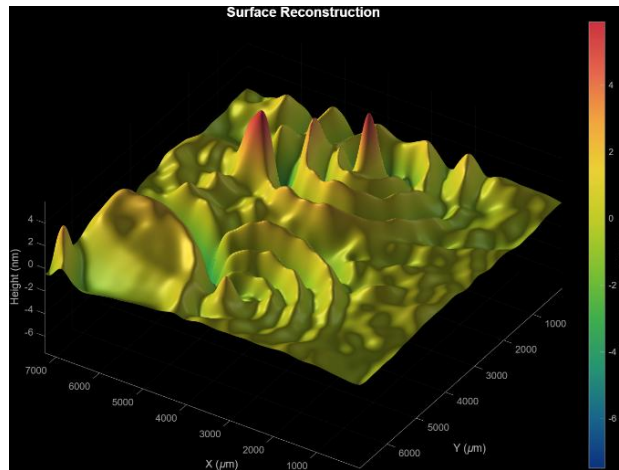


Figure 27 shows the three-dimensional reconstruction of the circular pattern. The reconstructed features closely match the reference image I_0 , with the concentric ring structure clearly preserved. Surface defects in the damaged LCD are also evident, demonstrating that both the imposed pattern and inherent screen irregularities contribute to the reconstructed surface.

Forked Phase

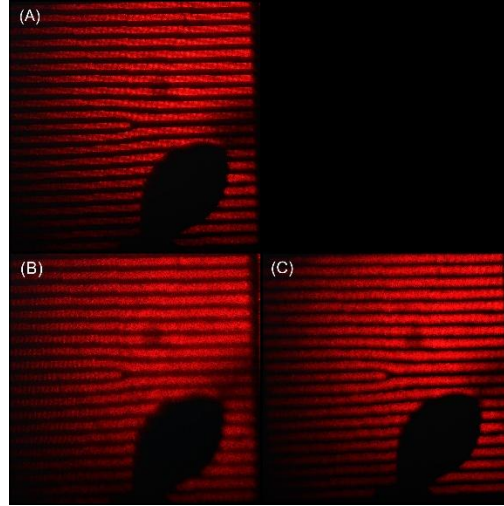


Figure 28 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a Forked Phase pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

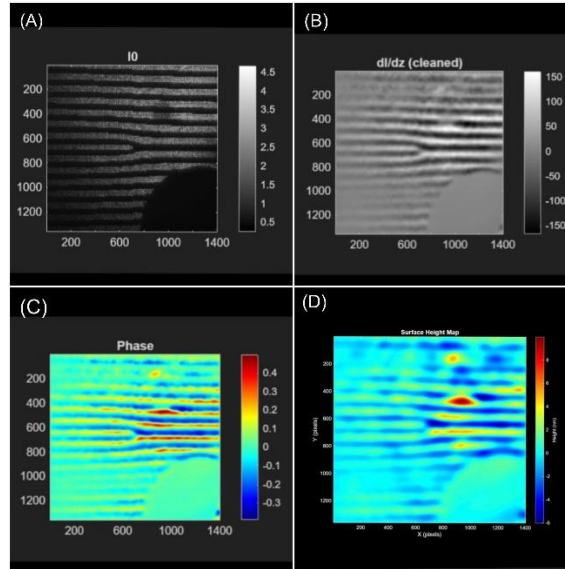


Figure 29 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

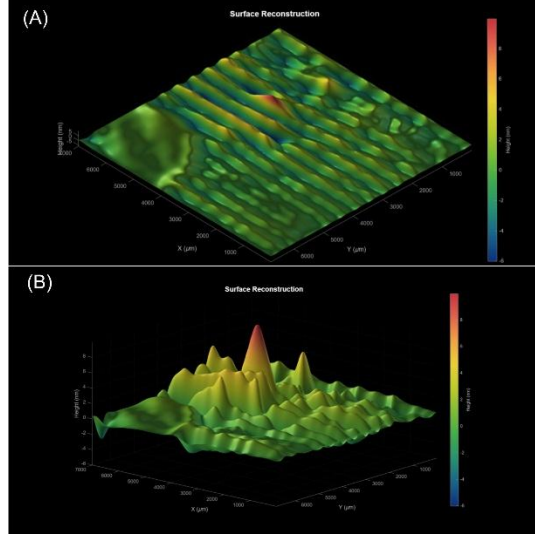


Figure 30 shows the three-dimensional reconstruction of the stripe pattern. The reconstructed features are consistent with the reference image I_0 , with the forked phase structure clearly preserved. Additionally, characteristics of the damaged LCD, including surface irregularities, are visible in the reconstruction.

In the next condition, the spatial light modulator (SLM) is turned on, but the code is adjusted to display no pattern. This setup is analogous to the initial condition shown in Figures 4-9, where the SLM was off and no image appeared on the LCD screen. In this case, the reconstruction produced a three-dimensional phase map corresponding to a blank display, which is consistent with expectations. Notably, the damaged region on the left side of the screen remains visible, confirming that the reconstruction captures physical imperfections in the device rather than applied modulation.

Additionally, a white filter was implemented in the code to reduce noise. This filtering improved the uniformity of the resulting images compared to earlier reconstructions without the filter, which exhibited more noticeable noise and fluctuations.

No pattern

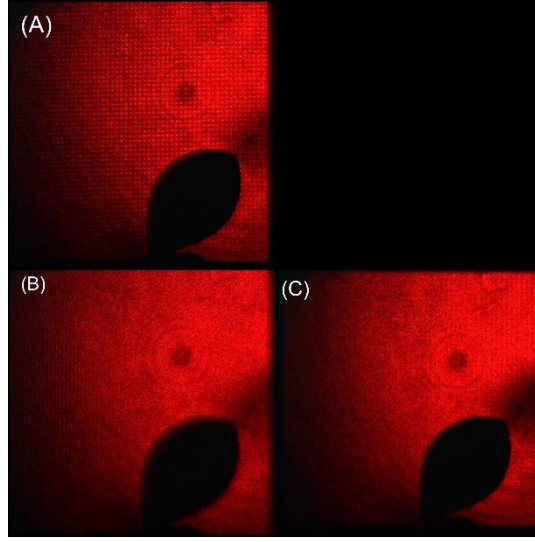


Figure 31 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying nothing just a white filter over the screen on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

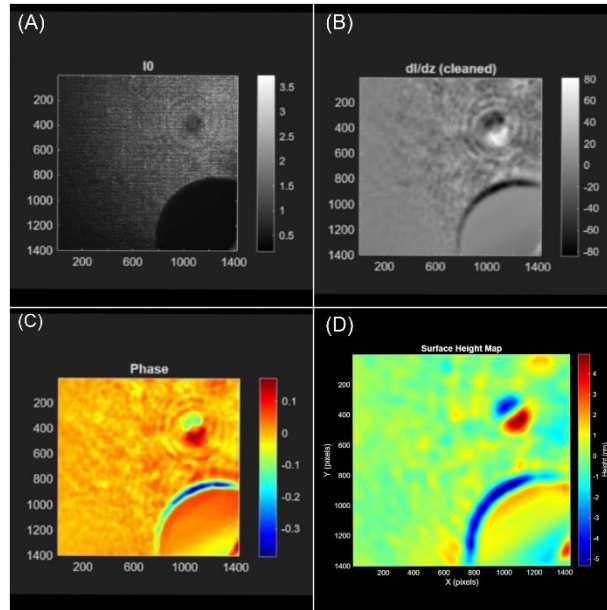


Figure 32 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

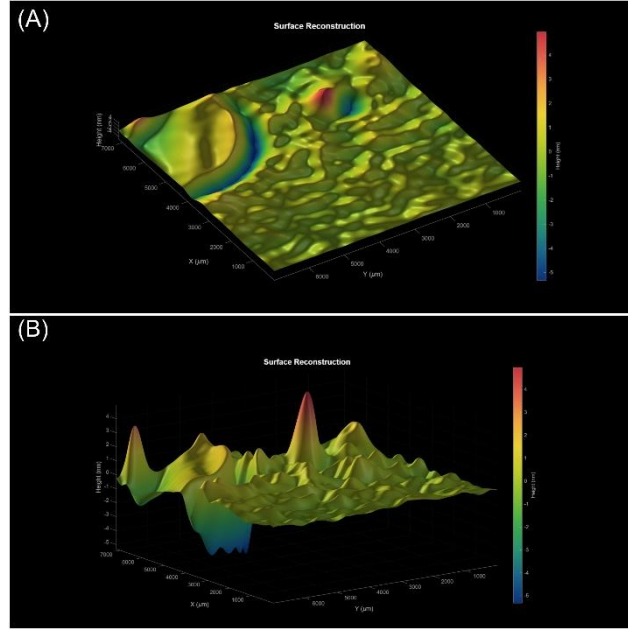


Figure 33 presents the three-dimensional phase reconstruction of the LCD screen under a no-pattern display condition, with (a) showing the top view and (b) the bottom view of the reconstructed surface. A pronounced localized feature is observed in the corner; corresponding to the damaged region of the LCD. Additionally, the central region exhibits a significant phase elevation, which is attributed to the point of laser incidence on the screen. These features confirm that the reconstruction is sensitive to both structural defects and localized optical interactions within the system.

IV. Discussions and Conclusions

Overall, the TIE reconstruction successfully recovered phase information from intensity-only measurements and produced a quantitative three-dimensional surface map of the LCD material. Despite the presence of minor low-frequency curvature and residual noise, the reconstructed structures correspond well with features observed in the raw intensity images, supporting the validity of the method.

The results demonstrate that phase nonuniformities and surface irregularities in the LCD can be effectively characterized without the need for interferometric techniques, highlighting the potential of the Transport of Intensity Equation as a low-cost and accessible tool for optical surface analysis and spatial light modulator evaluation.

However, the findings also reveal an important limitation: the TIE reconstruction is highly sensitive to spatial intensity variations. Under structured illumination, imposed patterns can be interpreted as phase variations, leading to artificial or exaggerated surface features in the reconstructed height maps. This indicates that care must be taken when applying TIE in systems with non-uniform or dynamically modulated illumination, as the recovered phase may not solely represent true physical geometry.

V. Future Works

To further validate and extend this work, additional experimental and computational developments will be pursued to better assess the accuracy, limitations, and overall performance of the low-cost LCD-based SLM system. While the current results demonstrate strong capability in capturing intensity variations and reconstructing surface features, further investigation is required to fully characterize the robustness of the method under varied conditions.

Future studies will explore the implementation of more advanced Transport of Intensity Equation (TIE) reconstruction techniques. In particular, the development of more sophisticated numerical solvers and regularization methods will be investigated to improve phase retrieval accuracy, reduce reconstruction artifacts, and enhance spatial resolution. These improvements are expected to provide more reliable phase maps, especially in regions with low signal-to-noise ratios or sharp discontinuities.

Experimentally, the SLM will be examined under a broader range of operating conditions, including uniform (no-pattern) illumination while the device is actively driven. This approach will help isolate the intrinsic phase response of the LCD from pattern-induced effects and provide deeper insight into the material and electronic behavior of the system.

A key technical challenge identified in this work is the conversion of standard image formats (e.g., PNG, JPEG, and PDF) into formats compatible with embedded systems such as Arduino-based controllers. Future efforts will focus on developing a robust pipeline for translating image data into executable code that can be directly used to drive the SLM. Enabling efficient image-to-code conversion will significantly expand the functionality of the system, allowing for dynamic and programmable phase modulation using arbitrary input patterns.

Additional work will also focus on improving noise reduction techniques and overall system stability to minimize wave-like artifacts observed in the reconstructed phase maps. This includes both hardware-level optimizations and computational filtering strategies.

Collectively, these advancements will provide a more comprehensive understanding of the optical behavior of low-cost SLM systems and significantly enhance the reliability and applicability of TIE-based phase reconstruction methods.

References

1. Zhao, S., Zhang, H., Liu, Y.-Q., Huang, L., Guan, R., Chen, Y., Cong, Y., Hong, Z., Wang, Z., & Liu, H. (2018). Spatial light modulator–based printing technologies for optical elements fabrication with different materials. *Optics and Lasers in Engineering*, 109, 140–148. <https://doi.org/10.1016/j.optlaseng.2018.04.019>
2. Márquez, A., & Lizana, Á. (2019). Special issue on liquid crystal on silicon devices: Modeling and advanced spatial light modulation applications. *Applied Sciences*, 9(15), 3049. <https://doi.org/10.3390/app9153049>
3. Matsuda, S., Shoda, M., Yoneda, N., Kumar, M., Watanabe, W., Murata, T., & Matoba, O. (2024). 3D fluorescence imaging through scattering medium using transport of intensity equation and iterative phase retrieval. *Optics Express*, 32, 10599–10617. <https://doi.org/10.1364/OE>.
4. Zuo, C., Feng, S., Huang, L., Tao, T., Yin, W., & Chen, Q. (2018). Phase shifting algorithms for fringe projection profilometry: A review. *Optics and Lasers in Engineering*, 109, 23–59. <https://doi.org/10.1016/j.optlaseng.2018.04.019>